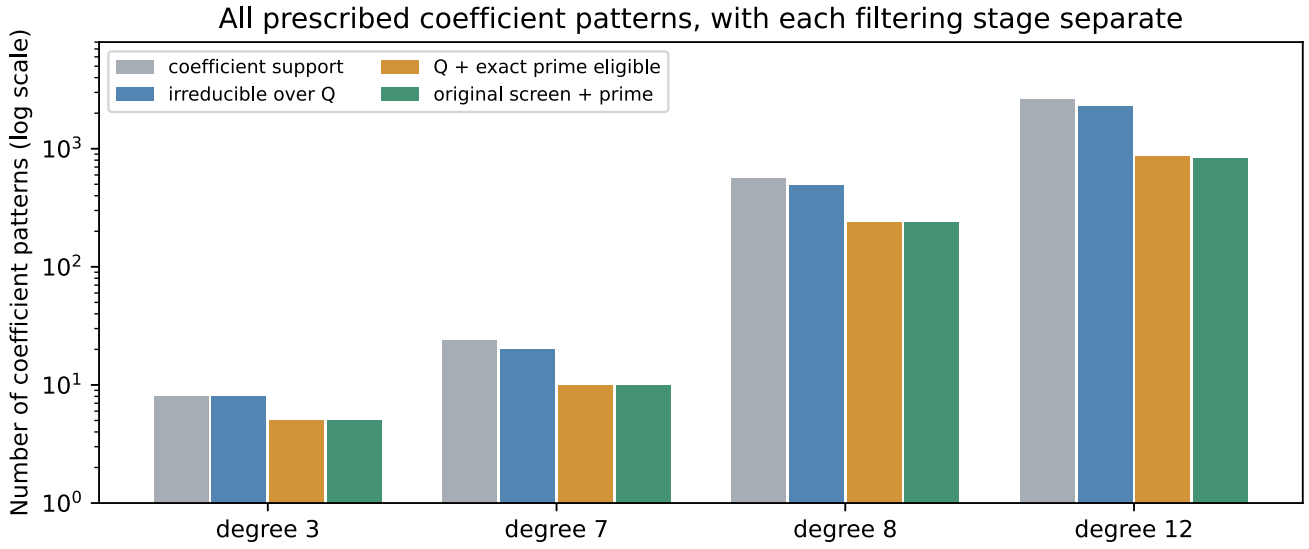
Complete finite sampler support

Observations only, 2026-10-18. EXP-ECDLP-00e491 completed both frozen trials; native statuses are output_validated. Built-in checks are not independent scientific review. Interpretation, relevance and usefulness calls remain with the owner.



d=3: 8 patterns, 8 Q-irreducible, 5 exact-prime eligible, 5 original-screen eligible; d=7: 24 patterns, 20 Q-irreducible, 10 exact-prime eligible, 10 original-screen eligible; d=8: 560 patterns, 489 Q-irreducible, 240 exact-prime eligible, 237 original-screen eligible; d=12: 2640 patterns, 2308 Q-irreducible, 858 exact-prime eligible, 837 original-screen eligible. These are exhaustive counts for the specified monic, constant +/-1, weight-three/five coefficient supports, k=4..12 and prime minimum 4096. Full QQ factorizations and all 29,088 prime records are retained.

What is worth inspecting: 3 degree-eight and 21 degree-twelve exact-prime-eligible, Q-irreducible patterns remain unresolved by the original modular irreducibility screen. Across all coefficient patterns, the corresponding Q-irreducible but unresolved counts are 7 and 80. A missing screen witness is not relabeled Q-reducible.

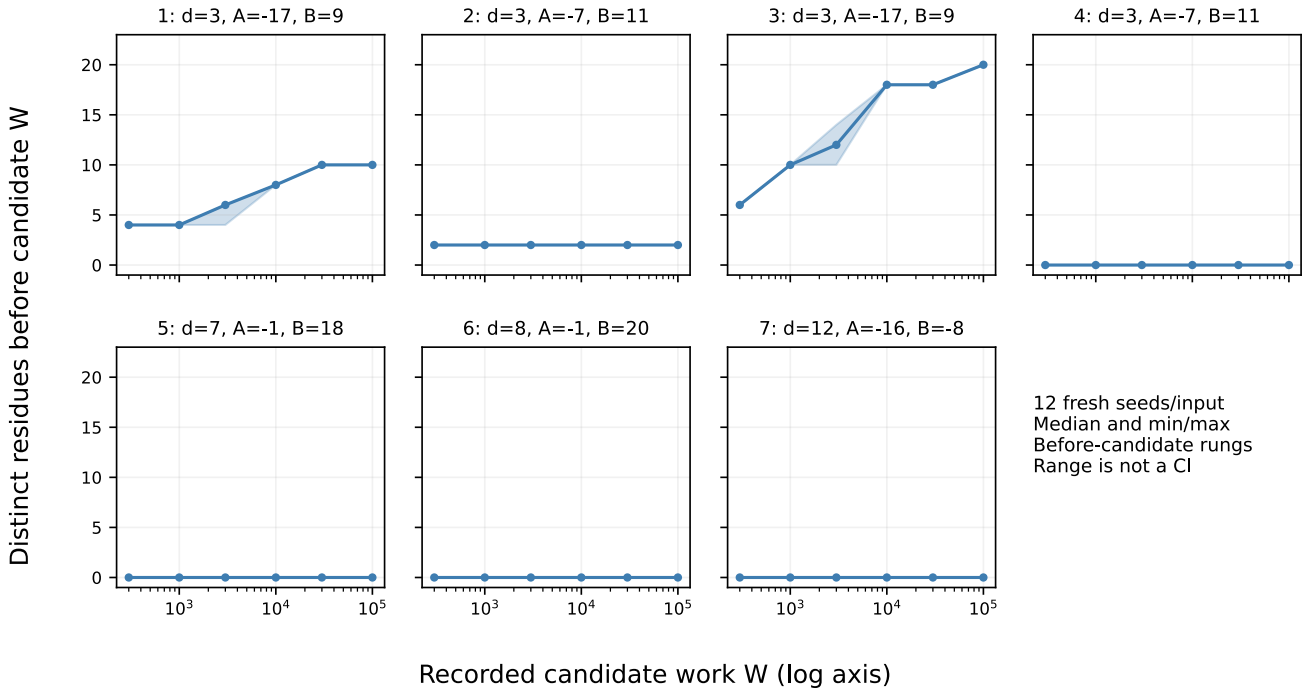
Both prime screens agree on all 29,088 eligible/ineligible value decisions. Every original positive irreducibility witness agrees with exact Q irreducibility. These agreements and unresolved cases are separate columns, not a uniformity or anomaly significance test.

Example unresolved eligible degree-eight pattern, coefficients low to high: [1, 0, -1, 0, -1, 0, -1, 0, 1]; first exact eligible k=7. All 24 eligible unresolved patterns and their exact factors are in observations.json. The source trial uses Sage 10.7 with proof-enabled primality; source-screen and exact classifications remain distinct.

Source RUN-ECDLP-ee0d23. Reproducible commands, seeds, hashes, execution-start commit, environment, child CPU and memory are in the sealed native records. No earlier G3 result or protocol was changed.

Fresh seeds with inputs held fixed

Seven fixed inputs, twelve fresh search-seed strings each: 84 fresh jobs, plus eleven legacy-seed controls and seven identical-seed replays. No field, curve or auxiliary-prime list is regenerated when a search seed changes. All 102 prescribed jobs completed.



input 1 (d=3): final D range 10..10; input 2 (d=3): final D range 2..2; input 3 (d=3): final D range 20..20; input 4 (d=3): final D range 0..0; input 5 (d=7): final D range 0..0; input 6 (d=8): final D range 0..0; input 7 (d=12): final D range 0..0. All endpoints match within each input's twelve fresh jobs, including ordinary zero-count cases. Intermediate-rung spread is retained: input 1: W=3000 D=4..6; input 3: W=3000 D=10..14.

Selection was fixed before launch: four exact archived inputs previously seen under seeds 1/3, plus the minimum canonical-JSON input key among archived seed-1 random_g cases at each degree 7,8,12. These seven conditional cases are not a random sample of all curves or fields. Distinct seed labels do not prove statistical independence.

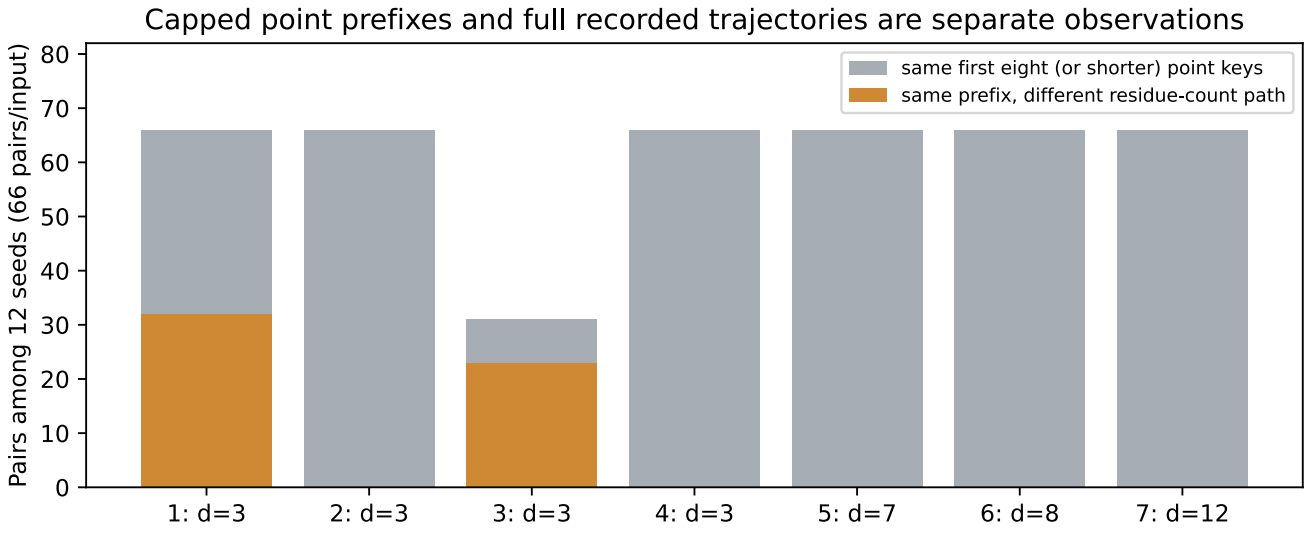
The plot shows the median and observed min/max of twelve counts at each recorded rung. The shaded ranges are not confidence intervals; no population variance, significance label or fitted scaling exponent is supplied. Point/residue counts do not identify full point sets.

Original checkpoints occur before testing candidate W. Final after-search snapshots are saved separately. Across all 102 jobs, residue-count differences between those last two boundaries were {"0": 102} (difference: job count). The boundary distinction is retained even when counts happen to match.

Source RUN-ECDLP-6bd7b4. Every unique discovered point, checkpoint and shell event is retained in the indexed instrument.tar.gz bundle, with no eight-point cap. Rejected candidates are not individually logged.

Prefixes, controls and provenance

Each input has 66 pairwise comparisons among twelve fresh jobs. These pairs reuse the same twelve jobs; they are not 66 independent samples. Point-key prefixes use coordinates, exclude event work counters and include the whole shorter list when fewer than eight points were found.



input 1: 66 same-prefix pairs, 32 of those with different D paths, 0 different full point-set pairs; input 2: 66 same-prefix pairs, 0 of those with different D paths, 0 different full point-set pairs; input 3: 31 same-prefix pairs, 23 of those with different D paths, 0 different full point-set pairs; input 4: 66 same-prefix pairs, 0 of those with different D paths, 0 different full point-set pairs; input 5: 66 same-prefix pairs, 0 of those with different D paths, 0 different full point-set pairs; input 6: 66 same-prefix pairs, 0 of those with different D paths, 0 different full point-set pairs; input 7: 66 same-prefix pairs, 0 of those with different D paths, 0 different full point-set pairs. Full per-input trajectories and pair counts are in observations.json; all zero-difference/control behavior is retained beside differences.

All eleven legacy controls reproduce every comparable original rung field. All seven identical-seed replays have byte-identical event streams. The separate checker verified all 480 discovered point events by exact Fraction arithmetic on the curve and recomputed residues; it also replayed all support records. No independent scientific review is claimed.

Producer timing (completed operations, not performance comparisons): support wall 14.762 s, child CPU 14.710 s, child peak RSS 245747712 bytes; panel wall 61.306 s, child CPU 61.392 s, child peak RSS 54910976 bytes. Checker time is separate from those producer metrics. Exact host/platform and launch provenance are preserved in native manifests.

All requested sampler follow-ups in this protocol are complete: larger-degree finite support, fixed-input interface, fresh seeds, legacy controls, identical-seed replay, uncapped traces and separate observation views. New questions to inspect are the structure of unresolved modular-screen cases and how discovery order changes while final counts agree. These are proposed follow-ups, not additional measurements or verdicts.

Raw records were published before this report. Certificate kind remains none; hypothesis/goal scientific conclusions and historical observations remain unchanged. Local coverage() retains a hardcoded publication_verified:false field; actual publication is evidenced by Git commits/hashes and the separate archive receipt, never inferred from that placeholder.